



Degree Project in Industrial Management and Data Science

First level, 15 hp

Comparative Analysis of Machine Learning and Deep Learning Models for Card Fraud Detection

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Comparative Analysis of Machine Learning and Deep Learning Models for Card Fraud Detection

Jämförande analys av maskininlärnings- och djupinlärningsmodeller för bedrägeriupptäckt i korttransaktioner

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Abstract—As the use of digital transaction grows, it brings with it an increased risk of fraud, affecting financial stability for both merchants and consumers. In 2021 alone, the global payment card fraud reached over \$ 32 billion, with estimations indicating potential losses of nearly \$ billion over the next decade. This study explores the effectiveness of different machine learning methods at detecting fraudulent transactions. Through a comparative analysis using a large data set from Kaggle.com, the study aims to enhance detection efficiency and accuracy, evaluating their performance using metrics such as accuracy, F1/score and ROC-AUC. The analysis not only focuses on technical evaluation of model effectiveness, but also considers how this aligns with real-world business goals and the bigger social and economic picture. Its about finding ways to use technology for good, making sure financial security is available to all. Being the base model, logistic regression (LR) got the lowest performance with 0.83 AUC-value compared to Extreme Gradient Boost (XGB) and Dense Neural Network (DNN) which got 0.93 and 0.84 AUC-value respectively. Same order were conducted for the F1-score with LR achieving a F1-score of 0.34, the XGB model 0.49 and the DNN model 0.38. The F1-score is particularly interesting where it considers coverage of both classes, which in broader terms means that the model considers both that valid transactions are allowed and that fraudulent transactions are prevented. The study determined that the XGBoost model exhibited superior performance among the various models tested. However, it's important for businesses to consider their operational constraints, particularly in terms of computing resources and budget allocations for cloud services or data storage. For organizations facing limitations in these areas, LightGBM presents an optimal choice. This model operates significantly faster than XGBoost and, while it offers a marginally lower performance, it still maintains a high level of accuracy. LightGBM's efficiency in handling large datasets with reduced computational demand makes it an ideal solution for businesses aiming to balance performance with cost-effectiveness.

Index Terms—Card Fraud Detection, Machine Learning, Deep Learning, Feature Engineering, Financial Security, Sustainable Development Goals, Data Imbalance, Fraud Analytics, Predictive Modeling.

A. Sammanfattning

I takt med att användningen av digitala transaktioner ökar, medför det även en ökad risk för bedrägerier, vilket påverkar den finansiella stabiliteten för både företag och konsumenter. Under 2021 uppgick det globala bedrägeriet med betalkort till över 32 miljarder dollar, och uppskattningar visar på potentiella förluster på nästan en miljard dollar under det kommande decenniet. Denna studie utforskar effektiviteten

av olika maskininlärningsmetoder för att upptäcka bedrägliga transaktioner. Genom en jämförande analys med en stor datamängd från Kaggle.com, syftar studien till att förbättra effektiviteten och noggrannheten i upptäckten, där prestanda utvärderas med hjälp av måttvärden som noggrannhet, F1-poäng och ROC-AUC. Analysen fokuserar inte bara på teknisk utvärdering av modellernas effektivitet utan beaktar även hur detta samspelar med verkliga affärs mål och större sociala och ekonomiska sammanhang. Det handlar om att hitta sätt att använda teknik för det gemensamma goda och att säkerställa att finansiell säkerhet är tillgänglig för alla.

Som basmodell hade logistisk regression (LR) den lägsta prestandan med ett AUC-värde på 0,85, jämfört med Extreme Gradient Boost (XGB) och Dense Neural Network (DNN), som fick 0,93 och 0,84 i AUC-värde respektive. Samma ordning genomfördes för F1-poängen där LR uppnådde en F1-poäng på 0,34, XGB-modellen 0,49 och DNN-modellen 0,38. F1-poängen är särskilt intressant eftersom den tar hänsyn till båda klasserna, vilket i ett bredare perspektiv innebär att modellen både tillåter giltiga transaktioner och förhindrar bedrägliga transaktioner. Studien fastställde att XGBoost-modellen uppvisade överlägsen prestanda bland de olika testade modellerna. Det är dock viktigt för företag att beakta sina operativa begränsningar, särskilt när det gäller datorresurser och budgetallokeringar för molntjänster eller datalagring. För organisationer som står inför begränsningar på dessa områden presenterar LightGBM ett optimalt val. Denna modell fungerar betydligt snabbare än XGBoost och, även om den erbjuder något lägre prestanda, bibehåller den fortfarande en hög nivå av noggrannhet. LightGBM effektivitet vid hantering av stora datamängder med minskad datorkraft gör den till en idealisk lösning för företag som strävar efter att balansera prestanda med kostnadseffektivitet.

I. INTRODUCTION

A. Background

IN the times of the digital economy's expansion, card transactions have become increasingly an integral part of our daily lives. However, this increase in online transactions has correspondingly led to an increase in card fraud incidents, posing a challenge on merchants, financial institutions, and consumers. A financial transaction becomes fraudulent when

there is a deliberate attempt to deceive and cause financial or personal gain. Fraudulent activities can range from unauthorized card usage, identity theft, to sophisticated schemes like synthetic identity fraud. The types of data relevant to fraud detection include transactional details such as the amount, date/time, and merchant category, alongside user-specific information like account history and location data. These datasets are critical as they help in identifying patterns and anomalies indicative of fraudulent behavior. The data set that is used in this study includes the many more features than the ones named above, which can be both good and bad for the models. According to the Nilson Report, payment card fraud worldwide exceeded \$32 billion dollars only in 2021, which was an increase of 14% compared to the year before. The report also mentions that the industry is projected to lose an accumulated \$398 billion in a span of ten years¹. These fraud incidents not only cause financial losses, but also insecurity among card users, hindering the potential and slowing down digital commerce. Addressing and solving card fraud incidents align directly with several Sustainable Development Goals (SDG) established by the United Nations. Combating card fraud supports responsible consumption patterns which ensure sustainable growth, aligning with SDG 12. Tackling card fraud drives innovation in the field of secure payment technologies, which enhances resilience in digital infrastructure for building a robust global economy, aligning with SDG 9. The fraud incidents disproportionately affect vulnerable populations. Protecting these individuals from financial fraud contributes to reducing financial insecurity and economic inclusion which align with SDG 1² and contribute significantly to societal well-being and economic health globally.

B. Problem

Detecting fraud within financial transactions presents two primary challenges: overfitting and bias. Overfitting occurs when a model learns the detail and noise in the training data to an extent that it negatively impacts the performance of the model on new data. Bias, either in the dataset or the model, can lead to skewed results, which disproportionately affect either the detection of fraud or the acceptance of legitimate transactions.

In fraud detection, there is often a trade-off between catching as many fraudulent transactions as possible (sensitivity) and allowing all legitimate transactions (specificity). The prioritization between these depends largely on the business's risk tolerance and operational requirements. For instance, a highly conservative approach that aims to minimize fraud might tolerate more false positives, leading to legitimate transactions being incorrectly flagged as fraudulent.

While existing fraud detection systems have achieved significant success in classifying and identifying fraudulent transactions, fraudsters continually evolve their tactics, which means detection systems must also adapt and evolve to catch new

types of fraudulent behaviour. Other important challenges to mention are class imbalance in data and the evolving nature of customer behaviour. All these challenges underscore the need of continuous improvement to catch up to the new challenges. This study aims to conduct a comprehensive comparison between different machine learning and deep learning methods to detect fraudulent transactions using data provided by Vesta Corporation, one of the forerunners in guaranteed e-commerce payment solutions³. The primary question this study aims to answer is reflect both technical and industrial managerial perspectives: **Given the continuous evolution of fraud tactics, how can financial institutions decide on the optimal model to invest in and develop further, based on the criteria of accuracy, speed, and complexity?**

The metrical accuracy is primarily measured using AUC ROC (Area Under the Receiving Operating Characteristic Curve) and F1-score, which are crucial metrics for assessing how well models perform on heavily imbalanced data - in this scenario a clear minority of fraudulent transactions. The additional criteria of speed and complexity are aligned with model training time, storage requirements, and computing power. These resources are vital for businesses utilizing paid and cloud services to process millions of transactions daily, as they translate directly into financial costs. Even marginal improvements in model efficiency can lead to significant savings in these resources, thereby reducing expenses for businesses. This study presents a mapping of what models and data transformation techniques should be used to align with specific business goals and requirements for their fraud detection systems.

C. Data

1) *Data Source*: All data used in this study was gathered and produced by Vesta Corp. and supplied on Kaggle.com, IEEE Computational Intelligence Society (IEEE-CIS) Fraud Detection competition. Kaggle is a data science competition platform and online community of data scientists and machine learning practitioners under Google LLC⁴. Vesta, the host of this competition, is a global transaction guarantee company for online purchases⁵.

2) *Selection of the Kaggle IEEE-CIS Dataset*: This dataset was chosen due to its comprehensive nature and the realistic simulation it offers for transactional fraud detection. The data includes a wide range of feature types commonly found in transactional systems, such as continuous amounts, categorical product codes, and engineered features, which are typical in real-world fraud detection tasks.

The choice also stems from the dataset's representation of common challenges in fraud detection, including class imbalance and the need for feature engineering to uncover subtle signs of fraudulent activities. These characteristics make the dataset not only a robust platform for developing and

¹The Nilson Report, "Payment card fraud losses reach \$32.34 billion," December 22, 2022, <https://www.globenewswire.com/news-release/2022/12/22/2578877/0/en/Payment-Card-Fraud-Losses-Reach-32-34-Billion.html>

²UN, "The 17 Goals" 2015, <https://www.globalgoals.org/goals/> [Accessed 06-03-2024]

³Vesta, "Learn More About Us — Vesta," 2023, <https://www.vesta.io/about-vesta> [Accessed 06-03-2024]

⁴Kaggle, "Kaggle: Your Machine Learning and Data Science Community," 2024, <https://www.kaggle.com/> [Accessed 06-03-2024]

⁵Vesta, "Learn More About Us — Vesta," 2023, <https://www.vesta.io/about-vesta> [Accessed 06-03-2024]

testing fraud detection models, but also ensure that the findings are generalizable to other transactional datasets typically seen in the industry.

3) *About The Data*: The data consists of two major subsets with approximately one million data points in total. The subsets are the training and the test data set. The only difference between the sets is that the training data set contains the additional Target Feature, "isFraud", which tells with certainty whether or not a data point (transaction) was fraudulent. During training the target feature is used to adjust the model parameters to the features of the training dataset. The model can then be applied to the new unseen test dataset, which contains the same features as the training dataset, for inference, estimating which datapoints are fraud or not. The target feature isFraud consists of two classes (labels), 0 for legitimate transactions (negative class) and 1 for fraudulent transactions (positive class) making this a binary classification problem.

The data also includes anonymized features engineered by Vesta (V1-V339). These features potentially encapsulate complex relationships related to the target variable and are crucial for enhancing the model's predictive accuracy without compromising user privacy. They represent large and complex feature engineering, hence why this study does not include further comprehensive feature engineering. (**kaggleData**)

We focus solely on the transaction dataset and deliberately exclude the identity-specific data set provided by Vesta. This is because integrating the two datasets would require Vesta specific data transformations that do not necessarily generalise well to other fraud classification problems. This choice reflects our study's primary objective—to compare various AI techniques, rather than striving to achieve the highest score on the leader board in this particular competition.

4) Brief overview of the features:

TransactionID	Transaction unique identifier
isFraud	Is or is not fraud (target feature)
TransactionsDT	Transaction Datetime
TransactionAmt	Transaction Amount [USD]
ProductCD	Product Codes (W, C, R, H, S)
card1 - card6	Product Codes (W, C, R, H, S)
addr1, addr2	Payment card info
dist1, dist2	Distance
P_emaildomain	Purchaser email domain
R_emaildomain	Receiver email domain
C1 - C14	Counting
D1 - D15	Timedelta
M1 - M9	Match
V1 - V339	Vesta engineered rich features

Categorical Features - Transaction

ProductCD	Product Codes (W, C, R, H, S)
card1 - card6	Product Codes (W, C, R, H, S)
addr1, addr2	Payment card info
P_emaildomain	Purchaser email domain
R_emaildomain	Receiver email domain
M1 - M9	Match

II. THEORY

The theory section provides brief explanations of the under-

lying models and data transformation algorithms included in this study, providing a basic but helpful insight.

A. Machine Learning Models

1) Standard Machine Learning Models:

2) *Logistic Regression (LR)*: Logistic Regression uses a logistic function to model a binary dependent variable. It calculates the probability of the default class (usually 1, the positive class) using the sigmoid function, $\sigma(z) = \frac{1}{1+e^{-z}}$, where z is a linear combination of the input features. The model parameters are optimized by maximizing the likelihood function, typically using gradient descent.

3) *Random Forest Classifier (RF)*: Random Forest constructs multiple shallow decision trees (weak learners) during training through bootstrap aggregating (bagging). Each tree is trained on a random subset of the data, and feature selection is randomized for splitting at each node. The final prediction is obtained by taking a majority vote from all individual trees in a democracy-like approach.

4) Gradient Boosting Machine Models (GBMs):

5) *XGBoost Classifier (XGB)*: XGBoost employs gradient boosting, which builds an ensemble of trees sequentially. Each tree corrects the errors of the previous one by fitting to the residuals. It uses a regularized objective to prevent overfitting, incorporating both the gradient (first derivative) and the Hessian (second derivative) of the loss function for more precise updates. The algorithm supports sparsity-aware learning and weighted quantile sketch for handling large datasets efficiently.

6) *CatBoost Classifier (CAT)*: CatBoost also uses gradient boosting on decision trees, but it is specifically designed to handle categorical features directly without preprocessing. It employs ordered boosting to reduce overfitting by using different permutations of the dataset for each iteration. Additionally, it uses a unique technique called oblivious trees, where the same splitting criterion is used across all nodes at the same depth. The internal data processing comes at a cost of efficiency.

7) *LightGBM Classifier (LGB)*: LightGBM uses gradient boosting with a focus on efficiency and scalability. It employs histogram-based algorithms for faster computation and uses leaf-wise (best-first) tree growth rather than level-wise (depth-wise) growth. This approach allows it to handle large datasets and high-dimensional data more effectively. LightGBM also supports exclusive feature bundling to reduce the number of features and gradient-based one-side sampling (GOSS) to focus on the most informative instances. This makes this model highly efficient.

B. Deep Learning Models

1) *Dense Neural Network Model (DNN)*: A Dense Neural Network (DNN) is a form of Multi Layer Perceptron (MLP) consisting of multiple layers of neurons, where each neuron in one layer is connected to every neuron in the next layer, thus dense. Each connection has an associated weight, which is adjusted during training. The network uses an activation

function (such as ReLU, sigmoid, or tanh) to introduce non-linearity akin to Logistic Regression. The weights are optimized using backpropagation, which involves computing the gradient of the loss function with respect to the weights and updating the weights using gradient descent or its variants.

C. Data

1) *Outlier Detection*: Cleaning the dataset of anomalous and non representative samples.

2) *Isolation Forest*: Isolation Forest is an ensemble method specifically designed for anomaly detection. It isolates observations by randomly selecting a feature and then randomly selecting a split value between the maximum and minimum values of the selected feature. The number of splits required to isolate a point is equivalent to the path length from the root node to the terminating node. Anomalies are those points with short average path lengths, as they are isolated quickly.

3) *Dimensionality Reduction*: Reducing the dataset size and complexity.

4) *Principal Component Analysis (PCA)*: Principal Component Analysis (PCA) is a linear dimensionality reduction technique that transforms the data into a new coordinate system. The first principal component is the direction of the maximum variance, the second principal component is orthogonal to the first and accounts for the next highest variance, and so on. This transformation reduces the dimensionality of the data while retaining as much variance as possible.

5) *Recursive Feature Elimination*: Recursive Feature Elimination (RFE) is a feature selection method that recursively removes the least important features and builds a model on the remaining features. It starts with all features and fits a model, then ranks the features by importance, eliminates the least important feature(s), and repeats the process until the desired number of features is reached.

6) *Class Imbalance*: Balancing importance of classifying both classes correct.

7) *Synthetic Minority-Oversampling Technique (SMOTE)*: SMOTE is an oversampling technique used to address class imbalance. It generates synthetic samples for the minority class by interpolating between existing minority samples. Specifically, for each minority class sample, SMOTE selects one or more of its nearest neighbors and creates new samples along the line segments joining the sample to its neighbors. This process increases the number of minority class samples in the dataset.

8) *L1 Regularisation (Lasso Regression)*: L1 Regularisation, also known as Lasso Regression, adds a penalty equal to the absolute value of the magnitude of coefficients to the loss function. The regularisation term is $\lambda \sum_i |w_i|$ where λ is a tuning parameter and w_i represent the model coefficients. This technique can shrink some coefficients to zero, effectively performing feature selection. This metric was used on Kaggle for evaluation on the leaderboard.

9) *L2 Regularisation (Ridge Regression)*: L2 Regularisation, also known as Ridge Regression, adds a penalty equal to the squared value of the coefficients to the loss function. The regularisation term is $\lambda \sum_i w_i^2$ where λ again is a tuning parameter and w_i represent the model coefficients. This technique

prevents overfitting by shrinking the coefficients but does not set them to zero.

10) *Data Splits*: The training data is divided into three sets: Training, Validation, and Testing, with an 80-10-10 split, for model evaluation. This partitioning is adopted because only the training dataset includes labels for local evaluation and empirical analysis. Fig. 1 illustrates this partitioning principle.

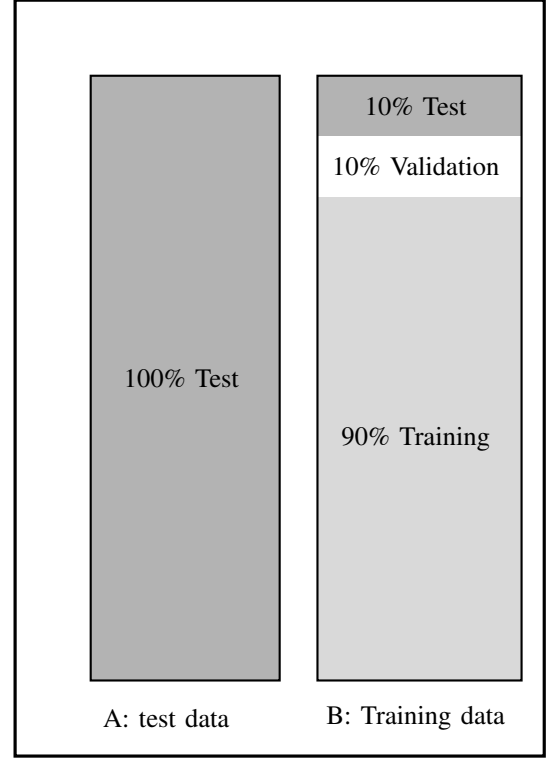


Fig. 1: Schematic representation of the dataset splits for model training and evaluation, displayed with horizontal labels. Panel A is dedicated entirely to testing. Panel B shows the split into training, validation, and testing sections with respective percentages.

D. Evaluation Metrics

See Fig. 2 for illustration of TP, FP, FN & TN metrics in a confusion matrix template.

1) *Area Under the Receiving Operating Characteristic Curve (AUC ROC)*: AUC ROC (Area Under the Receiver Operating Characteristic Curve) measures the area under the curve of the ROC plot, which depicts the trade-off between true positive rate (TPR) and false positive rate (FPR) at various fraud tolerance thresholds. $TPR = \frac{TP}{TP+FN}$ and $FPR = \frac{FP}{TN+FP}$. The FPR-TPR coordinates are plotted in a graph. As the decision boundary for fraud increases from 0% to 100% the number of correct fraud classifications (TPR) decreases but so also does the number of incorrect fraud classifications (FPR). Essentially tracking TPR against FPR as the model progresses from highly sensitive to highly specific in the fraud classification it makes. This is illustrated in Fig. 3.

2) *F1-Score*: F1-score is the harmonic mean of precision and recall, providing a single metric that balances false positives and false negatives and making it a robust metric for a holistic view of model performance that precision or recall misses individually. It is calculated as $f1_score = \frac{2 * recall * precision}{recall + precision}$ where $precision = \frac{TP}{TP + FP}$ and $recall = \frac{TP}{TP + FN}$.

3) *Model Benchmarking (Base Model)*: A base model serves as a minimum threshold for acceptable performance. Other models are then benchmarked against this base model to facilitate a comprehensive comparison.

		Actual Values	
		Positive (1)	Negative (0)
Predicted Values	Positive (1)	TP	FP
	Negative (0)	FN	TN

Fig. 2: Confusion Matrix illustrating how true positives (TP), false positives (FP), false negatives (FN) and true negatives (TN) are calculated for binary classification.

ROC-AUC was the measure used to evaluate models against each other in the kaggle competition this dataset was provided by. ROC curve is plotted with TPR along y-axis and FPR along x-axis as dependent variables on the decision boundary. AUC is the area under the ROC curve. TPR and FPR are given by the two following equations:

III. PREVIOUS RELEVANT WORK

Since the IEEE-CIS Fraud Detection competition on Kaggle ended five years ago but is still up as an open problem many old solutions and submissions exist. Previous top competitors have published their work and key findings from working live on this project some of which were implemented and used for this study. The inclusion of XGBoost, CatBoost and LightGBM models used were all inspired by previous submissions showing good results.⁶ In contrast to other submissions this study aims to lay out generalisable results and benchmark models and data techniques in a vacuum and not reach the top of Kaggle submissions as that requires the use of techniques highly specific to just this problem and not necessarily other.⁷ Several studies have worked on

⁶2nd Solution, CPMP View” 2019, <https://www.kaggle.com/competitions/ieee-fraud-detection/discussion/111321> [Accessed 19-02-2024]

⁷1st Place Solution - Part 1” 2019, <https://www.kaggle.com/c/ieee-fraud-detection/discussion/111284> [Accessed 19-02-2024]

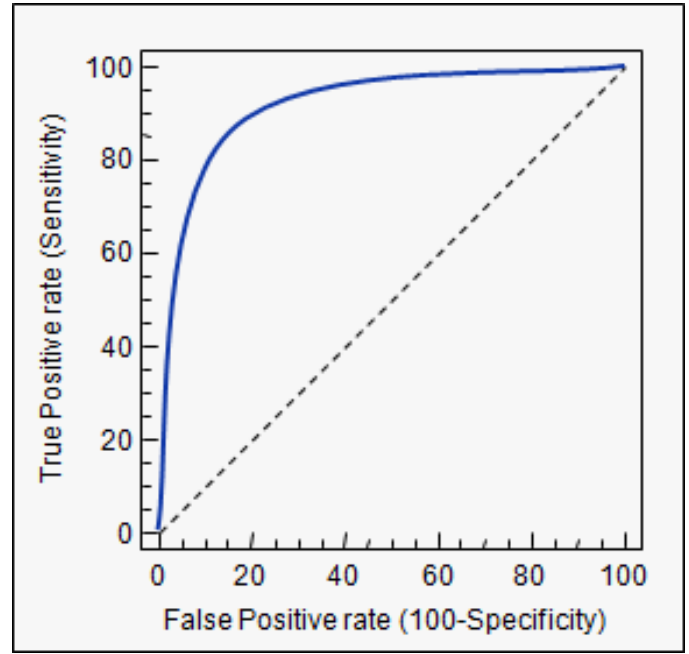


Fig. 3: Receiver Operating Characteristic (ROC) curve. The area under the curve (AUC) values provide insight into the accuracy of each model, with closer to 1.0 indicating higher predictive accuracy.

the challenge to detect fraudulent transactions and suggested different approaches to solve the problem. Their remarkable efforts have provided invaluable insights that paved the way for effective fraud detection. In 2017, Dr. Dal Pozzolo introduced a realistic modelling approach and a novel learning strategy for fraud detection. Pozzolo emphasised the importance of addressing concept drift and classification delay in designing fraud detection systems. He proposed a learning strategy that effectively managed these challenges, achieving high detection accuracy AUC metric around 0.85 (Dal Pozzolo et al., 2017). Building upon these foundations, in 2020, Dr. Ileberi and colleagues leveraged the power of Genetic Algorithms (GA) to meticulously select relevant features from vast transaction data. That was the core new idea in Dr. Ileberi and his colleagues research in my opinion. By combining these features with sophisticated machine learning algorithms, they reached the pinnacle of fraud detection accuracy, surpassing 99.98% (Ileberi et al., 2022). More recently, in 2022, Bandyopadhyay and their team embraced the capabilities of Recurrent Neural Networks (RNNs) to detect fraud in real-time. Their groundbreaking model demonstrated exceptional accuracy and resilience against the ever-changing nature of fraudulent activities. The result from the module they built shows that the model performs significantly well in terms of fraud transaction detection, scoring 99.87 in accuracy metric and 0.99 in F1 score (Bandyopadhyay, 2020).

Furthermore, a proposal for online analysis and identification of fraudulent financial transactions by researchers highlighted the inadequacy of predefined patterns in identifying evolving fraud methods. They suggested a hybrid approach using unsupervised outlier detection algorithms and Visual

Analytics (VA) to reduce false positives. Their framework, applied to a Brazilian financial institution handling over 30 million daily transactions, showed potential in outperforming current methods (Araujo Lima Torres, 2020).

Additionally, a 2017 study on deep learning for detecting fraud in credit card transactions evaluated various deep learning topologies, including artificial neural networks and long short-term memory (LSTM) networks. This research highlighted the importance of model parameters such as the number of neurons in the hidden layers and demonstrated that deep learning could provide results comparable to traditional methods like Gradient Boosted Trees and Logistic Regression. Using a high-performance distributed cloud computing environment, this study addressed common issues such as class imbalance and scalability, offering a comprehensive guide to parameter tuning for optimal fraud detection performance (Kumar Sadineni, 2020).

Lastly, research on detecting fraudulent credit card transactions using techniques such as ANN, Decision Trees, SVM, Logistic Regression, and Random Forest was conducted. The Random Forest algorithm and ANN showed exceptional performance, achieving accuracies of 99.21% and 99.92%, respectively. This highlights the importance of powerful techniques to manage the challenges of imbalanced data and the rapid production of transaction data (Abhimanyu, 2018).

These studies collectively underline the evolving landscape of fraud detection and the critical role of machine learning and deep learning models in developing robust and adaptable solutions. Our study builds on these insights, conducting a comparative analysis of various models to establish generalised benchmarks for card fraud detection.

IV. METHODOLOGY

The study was conducted in two phases.

The first phase involved a comparative empirical analysis of various machine learning models against Logistic Regression, which served as the baseline model. All models were trained on the same minimally preprocessed dataset. In this phase, the dataset remained constant while the models varied.

In the second phase, XGBoost, identified as the top-performing model balanced both in terms of performance, efficiency and simplicity, was used to evaluate the impact of different data transformation techniques. In this phase, the model was kept constant while the data transformations varied.

A. Hardware Specification

The computational experiments for this study were performed on a single hardware platform (Oskar's Mac book) to maintain consistency in performance evaluation across all tests. The specifications of the system used are detailed below:

- **Processor:** Apple M1 chip 8 core
- **RAM:** 8 GB
- **GPU:** Apple M1 8-Core GPU
- **Storage:** 256GB SSD
- **Operating System:** MacOS version 13.5.2

B. Exploratory Data Analysis (EDA)

EDA played a critical role in this study, was a very crucial part of this study. The abundance of features presented a challenge in determining the relevance of each feature to the models. Two particularly valuable EDA techniques were heat maps and correlation analysis. Heat maps provided a rapid visualization of feature characteristics, such as the proportion of missing values (NaN), as illustrated with figure 2 in the appendix, accompanied in the end of the report.

Correlation analysis maps further elucidated the relationships between data points within the dataset, also shown in the figure 3 in the appendix. These two EDA:s, along with others, facilitated visualization and simplified decision-making during the feature engineering process.

C. Feature encoding

Our approach to preparing the dataset for modeling involved a series of transformations tailored to handle various data types efficiently, such as ordinal categorical, data with only 2 unique values and nominal categorical data. To ensure robustness and efficiency in our models, we adopted strategies to manage high cardinality features effectively.

- **Binary Encoding:** Boolean columns were converted to binary format to simplify the input data structure and reduce memory usage. 1 or 0, and missing values were encoded as integers, facilitating a more efficient computational process and optimizing storage requirements.
- **Categorical Encoding and Analysis:** We utilized pivot tables to analyze categorical variables based on their association with the target variable, *isFraud*. This approach enabled us to discern the distribution of fraudulent activities across different categorical levels and apply hierarchical encoding based on the observed fraud rates. This method of encoding helps in capturing the underlying patterns of fraud within categorical data, improving the model's ability to detect fraudulent transactions.
- **Risk Group Encoding:** We categorized categorical features into risk groups based on their fraud occurrence rates. By classifying features into low, medium, and high-risk categories, we enhanced the model's interpretability and its capacity to prioritize resources towards more suspicious activities.
- **Preprocessing Pipeline:** Our preprocessing pipeline incorporated all these encoding strategies mentioned above while also addressing other crucial aspects such as the removal of redundant features and the management of missing values. This standardized preprocessing ensured that both training and test datasets were optimally prepared, maintaining consistency across all data handling stages.

D. Feature Selection (FS)

To avoid overfitting, minimal feature selection was employed using correlation analysis. Highly multicollinear and low-impact features were removed. Features with a correlation of 90.0% or more with others, or less than 0.5% with the target

feature (isFraud), were deemed problematic and were dropped, resulting in the removal of 101 features. This reduced overfitting, eliminated redundant features, and improved training times by reducing dimensionality of the dataset.

E. Choosing the Optimal Model for Data Transformation Techniques

After evaluating various models on the minimally transformed dataset, we selected one model to keep fixed for testing further data optimization. The candidate model was chosen based on its superior performance, in first hand high Kaggle AUC ROC scores but also weighing in training time for efficient testing. XGB showed a good balance of performant AUC ROC scores while keeping low latency for training and inference.

F. Keeping the Model Fixed - XGB

From previous testing of the XGBModel on the original dataset 194 training rounds (tree estimators) was found to be the optimal number. For each data transformation technique tested 194 estimators was kept as the value for the hyper parameter "n_estimators".

G. Feature Engineering (FE)

As feature engineering (FE) is highly dataset specific and may not be transferrable to other problems FE was excluded. The Vesta Engineered features already provided rich insights and constituted some of the most important features for XGBoost.

1) *Outlier Detection and Removal*: Isolation Forest (ISOF) was used to remove non representative datapoints from the training dataset while keeping the testing dataset intact. An IsolationForest model was trained on the training dataset with "auto" detection for the contamination hyperparameter. This resulted in 0.3 percentage of the training dataset to be classified as anomalous and dropped before training.

2) *Dimensionality Reduction*: Recursive Feature Elimination (RFE) was employed using the XGBoost models feature importance values from training it on the data. Iteratively the model was trained and features with less than 0.001 in feature importance was removed resulting in a total 101 features to be dropped from the dataset, more than a third of all of the features. The recursion of this method continued until no more features of less importance was identified. This provides large computational benefits increasing speed with the hope of little loss in accuracy. RFE also has the potential to remove redundant and distracting features causing overfitting, essentially increasing accuracy as well. Principal Component Analysis (PCA): For the vesta features V1-V339 PCA may prove beneficial as a means to reduce the dimensionality without major loss of information (explainable variance). These features are already highly anonymised and all numerical meaning the features have little to no interpretable power as is. Testing is required to know whether the computational speed gained outweighs the information loss from PCA. The dataset was reduced down from the initial 296 features to 50 with a major compression of information.

H. Balancing The Dataset

Synthetic Minority Oversampling Technique (SMOTE) is a method to balance an imbalanced dataset to avoid class bias from training and during model inference. The target feature 'isFraud' in the dataset provided by Vesta is binary and thus has two classes, 1 for transactions that are fraudulent and 0 for valid transactions. Fraudulent transactions are heavily underrepresented with only 20663 fraud records and 569877 valid records meaning only about 3.5% of all records are fraudulent. Thus techniques like SMOTE will be considered and used in combination with undersampling techniques to balance out the dataset.

SMOTE synthesises new samples in between the records of nearest neighbours meaning both old and new points will be bundled densely together. This has some caveats:

SMOTE has a tendency to cause overfitting due to synthetic datapoints being added locally to the other minority class datapoints. To not oversample excessively the current rate of fraud datapoints of 3.5% was only doubled to 7% to not make the model overly focused on the positive class and cause large losses in performance for the negative class. By this moderate increase the model have more datapoints to adjust to without creating a disproportional representation of the positive class.

Creating synthetic data always affects decision boundaries, especially in areas where data points were previously sparse contributing to the overfitting in those areas. Thus regularisation techniques will be crucial.

1) *Regularisation*: To avoid overfitting and impact of redundant features regularisation was added to the models. Regularisation effectively downscales unimpactful weights by adding the sum of weights as a penalty term to the loss function used during training. A near zero weight means the corresponding feature is of little importance for the outcome of each prediction. L1 and L2 regularisation was added sequentially each tested with hyper parameter value 10 and 100. Optimal regularisation was found at L1=10, L2=100.

2) *Combining the Data Transformation Techniques*: After testing and assessing each data transformation technique individually all techniques with positive impact on the Kaggle submission AUC ROC score were used together as a way of seeing how they worked collaboratively. The interaction of multiple techniques is not linear and simply additive and some of the techniques employed had negative synergies. The data transformation techniques was applied in the order of feature selection (RFE), then outlier removal (ISOF), then class balancing (SMOTE) and then lastly regularisation (L1=10 and L2=100). To assess how the model performs on this new transformed dataset the number of training rounds (n_estimators) was increased to 500 to let the model learn deeper patterns of the data without the risk of overfitting from the previous original dataset. This lead to the gratest increase in AUC ROC performance on Kaggle.

V. RESULT

The results start with the classification report for LR, which is our base model. It is meant to be a reference point which is used to benchmark the other modules.

A. Logistic Regression (LR)

TABLE I: Classification Report for Logistic Regression

	Precision	Recall	F1	Support
0	0.97	1.00	0.98	56841
1	0.67	0.23	0.34	2213
Accuracy			0.97	59054
Macro avg	0.82	0.61	0.66	59054
Weighted avg	0.96	0.97	0.96	59054
ROC AUC for Logistic Regression:	0.8405			

The logistic regression model constitutes the base model.

For the negative class (0, notFraud) the logistic regression model achieved a precision score of 97% and a recall score of 1.00%. This means that out of all of the transactions the model predicted as negative (TN+FN) (notFraud) the model were correct in 97% of its guesses and out of all of the actual negative transactions (TN+FP) the model were correct in almost 100% of its guesses. This relates to credit worthy customers with valid transactions. The overall f1-score for the negative class was 0.98 (base reference for other models).

For the positive class (1, isFraud) the logistic regression model achieved a precision score of 67% and a recall score of 23%. This means that out of all of the transactions the model predicted as positive (TP+FP) (isFraud) the model were correct in 67% of its guesses and out of all of the actual positive transactions (TP+FN) the model were correct in 23% of its guesses. This relates to dishonest customers with fraudulent transactions. The overall f1-score for the positive class was 0.34 (base reference for other models).

The LR model achieved an AUC ROC score of 0.8405.

B. Random Forest (RF)

TABLE II: Classification Report for Random Forest

	Precision	Recall	F1	Support
0	0.97	1.00	0.99	56841
1	0.83	0.32	0.46	2213
Accuracy			0.97	59054
Macro avg	0.90	0.66	0.72	59054
Weighted avg	0.97	0.97	0.97	59054
ROC AUC for Random Forest:	0.8887			

The Random Forest (RF) model performed better than the LR model both in terms of precision and recall for the positive class scoring a total f1-score of 0.46, +0.12 as compared with LR. The negative class was slightly higher for the overall f1-score of 0.99. AUC was also higher than LR at 0.8887.

C. Extreme Gradient Boost (XGB)

XGBoost (XGB) also scored higher than LR on both precision and recall but less precision and more recall than RF, totaling an f1-score of 0.49. The XGB AUC score was considerably higher than the LR AUC at 0.9044.

CatBoost (CAT) scored also higher on both precision and recall than LR totaling an f1-score of 0.48, slightly lower than

TABLE III: Classification Report for Extreme Gradient Boost

	Precision	Recall	F1	Support
0	0.98	1.00	0.99	56995
1	0.79	0.36	0.49	2059
Accuracy			0.97	59054
Macro avg	0.90	0.66	0.74	59054
Weighted avg	0.97	0.97	0.97	59054
ROC AUC for Extreme Gradient Boost:	0.9044			

TABLE IV: Classification Report for Categorical Boost

	Precision	Recall	F1	Support
0	0.97	1.00	0.99	56841
1	0.81	0.34	0.48	2213
Accuracy			0.97	59054
Macro avg	0.89	0.67	0.73	59054
Weighted avg	0.97	0.97	0.97	59054
ROC AUC for Extreme Gradient Boost:	0.9070			

XGB. CAT did however score the highest AUC out of all the models at 0.9070.

TABLE V: Classification Report for Light Gradient Boosting Machine

	Precision	Recall	F1	Support
0	0.98	0.99	0.98	56841
1	0.72	0.35	0.47	2213
Accuracy			0.97	59054
Macro avg	0.85	0.67	0.73	59054
Weighted avg	0.97	0.97	0.97	59054
ROC AUC for Extreme Gradient Boost:	0.9034			

LightGBM (LGB) achieved also higher scores for both precision and recall than LR totaling an f1-score of 0.47, just above RF at 0.46. LGB AUC at 0.9034 was also higher than RF but lower than the other GMBs (XGB CAT)

D. Dense Neural Network - (DNN)

TABLE VI: Classification Report for Dense Neural Network

	Precision	Recall	F1	Support
0	0.97	1.00	0.97	56995
1	0.74	0.25	0.38	2059
Accuracy			0.97	59054
Macro avg	0.85	0.62	0.68	59054
Weighted avg	0.96	0.97	0.96	59054
ROC AUC for DNN:	0.8478			

The Dense Neural Network (DNN) surprisingly barely outperformed the LR model with slightly higher precision and recall totaling an f1-score of 0.38. DNN AUC was also barely higher than LR at 0.8478.

E. Model Performance Overview on Kaggle.com

The table above details the AUC scores obtained from leaderboards on Kaggle. These scores represent the model's

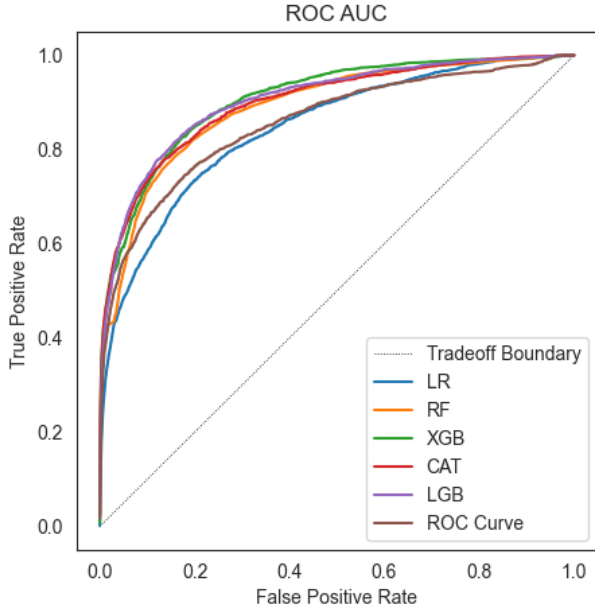


Fig. 4: Receiver Operating Characteristic (ROC) curve for all of the models during the parameter tuning and testing phase.

TABLE VII: Summary of Model Performances for the models

Metric	LR	RF	XGB
Rounds	400 (iter)	400 (est)	194 (stop)
Training Time (s)	324.25	506.1	21.5
Time/Round (s)	0.81	1.27	0.11
Inference Time (s)	0.4	33.0	0.9
AUC VSCode	0.8405	0.8865	0.9044
F1-score	0.34	0.46	0.49
AUC Kaggle	0.8317	0.9097	0.9116
Metric	CAT	LGB	DNN
Rounds	368 (stop)	400 (est)	14 (stop)
Training Time (s)	136.7	13.0	151.4
Time/Round (s)	0.37	0.03	10.81
Inference Time (s)	0.5	2.8	7.2
AUC VSCode	0.9069	0.9033	0.8478
F1-score	0.48	0.47	0.38
AUC Kaggle	0.9159	0.8945	0.8059

ability to classify transactions accurately under the receiver operating characteristic curve analysis, with a score closer to 1 indicating better performance.

F. further data optimization with XGB

The choosed model of

G. Analysis of Results

The RF model exhibited the best performance among the traditional machine learning techniques, with significant improvement seen in AUC scores. This model's ability to handle the imbalanced nature of the data effectively contributed to its high scores.

In contrast, the CNN model, typically robust in image processing tasks, showed lower effectiveness in this context,

TABLE VIII: Impact of Different Data Transforming Methods on AUC Kaggle using XGB

Data Transforming Method	AUC Kaggle	Delta
Original	0.911621	0
ISOF	0.915054	0.003433
RFE	0.912803	0.001182
PCA	0.821094	-0.090527
SMOTE	0.914068	0.002447
L1 (Lasso, alpha)	0.927239	0.015618
L2 (Ridge, lambda)	0.923882	0.012261
Combined techniques (est. = 194)	0.923527	0.011906
Combined technique (est. = 500)	0.929768	0.018147

likely due to the abstract nature of transaction data which lacks the spatial features that CNNs capitalize on.

The Dense Neural Network improved slightly over CNN, suggesting that for tabular data like that used in fraud detection, simpler dense architectures might outperform more complex structures like CNNs.

XGBoost and CatBoost, both gradient boosting models, performed exceptionally well, with CatBoost slightly leading. These models are known for their high accuracy and ability to handle various data types efficiently, which is critical in handling complex features inherent in financial transactions. But XGBoost achieved almost the

VI. DISCUSSION

The evaluation of various machine learning models on the fraud detection dataset reveals significant insights into the performance and effectiveness of each model. Starting with Logistic Regression (LR) as our base model, which served as a benchmark, we observed distinct performance metrics across different models, notably in terms of precision, recall, F1-score, and ROC AUC.

A. Analysis of Model Performance

The Logistic Regression model, while basic, provided a foundational understanding of the dataset's characteristics. It achieved high precision for the negative class (non-fraudulent transactions), indicating strong reliability in identifying legitimate transactions. However, its performance on the positive class (fraudulent transactions) was less impressive, which is a critical aspect for fraud detection systems aiming to minimize false negatives.

The advancement to more complex models like Random Forest and Extreme Gradient Boosting (XGB) resulted in improved detection of fraudulent transactions, as evidenced by their higher F1-scores and ROC AUC values. These models better balanced the trade-off between sensitivity and specificity, reducing the risk of false negatives while maintaining high accuracy overall. Notably, XGB and Random Forest exhibited enhanced abilities to generalize from the training data, likely due to their robust handling of imbalanced datasets and complex feature interactions.

B. Further Optimization

The decision to further optimize the XGBoost model was based on its superior performance metrics and its ability

to handle large-scale data effectively. The model performed slightly behind CatBoost, but was still chosen to develop further due to its superiority when it comes to time take for taring. This decision reflects the main question for the study, meaning that criteria like time and complexity also to be considered. Additional data transformation techniques like Isolation Forest (ISOF), Recursive Feature Elimination (RFE), and regularization methods (L1 and L2) were explored to enhance the model's performance. Each method brought improvements in AUC scores on Kaggle, demonstrating the potential for fine-tuning machine learning models to achieve higher predictive accuracy.

C. Prioritization of Model Metrics:

In fraud detection, the balance between sensitivity and specificity is critical. Sensitivity prioritizes capturing as many fraudulent transactions as possible, potentially at the expense of falsely denying valid transactions. This is particularly important in contexts where failing to detect fraud can have severe consequences, similar to over-classifying individuals in disease screening to prevent untreated cases. Conversely, specificity focuses on ensuring valid transactions are not flagged as fraudulent, aligning with principles like the legal standard of 'innocent until proven guilty.' Deciding which metric to prioritize often depends on the business's operational goals and the potential impact of false positives versus false negatives.

D. Strategic Considerations for businesses in Model and Data Transforming Selection

Businesses must carefully select their model training strategies based on their operational needs and the types of fraud risks they most frequently encounter. For instance, businesses seeking high sensitivity, which involves detecting as many fraudulent transactions as possible, should focus on implementing Synthetic Minority Over-sampling Technique (SMOTE). This method enhances the detection of fraud by artificially balancing the dataset, thereby increasing the model's ability to identify fraudulent transactions often missed due to data imbalance.

However, while SMOTE enhances sensitivity, it can reduce specificity—meaning there's an increased likelihood of false positives, where legitimate transactions are incorrectly flagged as fraudulent. Thus, businesses prioritizing high specificity, where the goal is to correctly validate as many legitimate transactions as possible, should generally avoid using SMOTE. This approach minimizes disruption to genuine transactions and improves customer satisfaction by reducing unnecessary fraud alerts.

Further Considerations:

- **Generalizability:** To build models that perform well across various scenarios and datasets, businesses should consider regularization techniques like L1 and L2 regularization, which prevent overfitting by penalizing large coefficients in the model. Techniques such as Recursive Feature Elimination (RFE) and Isolation Forest (ISOF) also contribute to model robustness. RFE reduces model

complexity by iteratively removing features that contribute the least to the prediction power, while ISOF helps in handling outliers, which improves the model's generalization across different datasets.

- **Low Latency:** For applications requiring fast decision-making, such as real-time transaction processing, models that offer quick training and inference times are essential. Light Gradient Boosting Machine (LGB) and PCA (Principal Component Analysis) are notable for their efficiency. LGB optimizes both speed and performance for large datasets, and PCA reduces dimensionality, which speeds up model training without sacrificing too much accuracy.
- **High Interpretability:** In scenarios where understanding the decision-making process of the model is crucial—for instance, in complying with regulatory requirements—Logistic Regression (LR) and Random Forest (RF) are preferred. These models provide clear insights into how input features affect predictions, making them valuable for audits and regulatory reviews.
- **Most Performant:** For outright performance, measured in terms of accuracy, precision, and recall, the Categorical Boosting (CAT) model often stands out. It combines the advantages of gradient boosting with categorical feature support, making it particularly powerful in handling diverse datasets.
- **Balanced Approach:** If a balance between performance and model complexity is required, Extreme Gradient Boosting (XGB) provides an excellent middle ground. It offers robust performance across various metrics without the extensive parameter tuning required by more complex models.

In conclusion, selecting the right model and preprocessing techniques depends largely on the specific needs of the business and the characteristics of the data. Each method brings its strengths and trade-offs, and the optimal choice will depend on the desired balance between sensitivity, specificity, speed, interpretability, and overall performance.

E. Future Research Directions

Future studies can build on this work by exploring several avenues:

- **Integration of Transactional Context:** Incorporating more contextual data about transactions, such as time patterns and behavioral analytics, could enhance the model's sensitivity to emerging fraud tactics.
- **Deep Learning Approaches:** While traditional machine learning models have shown robust performance, deep learning architectures could be explored further for their ability to capture non-linear and complex patterns in large datasets.
- **Cross-Dataset Validation:** Applying the optimized models to different datasets or across various financial platforms can validate their generalizability and robustness against diverse types of fraud.
- **Real-Time Processing:** Enhancing models to operate effectively in real-time environments, ensuring they can

make quick decisions without sacrificing accuracy.

VII. AUTHORS' CONTRIBUTIONS

Both authors, Oskar Wallberg and Yamen Alkattab, made substantial contributions to every aspect of this study, ensuring a comprehensive and collaborative effort throughout the project. Yamen Alkattab focused primarily on the literature study, data visualising & pre-processing functions and prepared the report following the IEEE formatting guidelines. Oskar Wallberg, on the other hand, took the lead in developing the AI models, feature engineering and fine tuning the features.

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VIII. APPENDIX

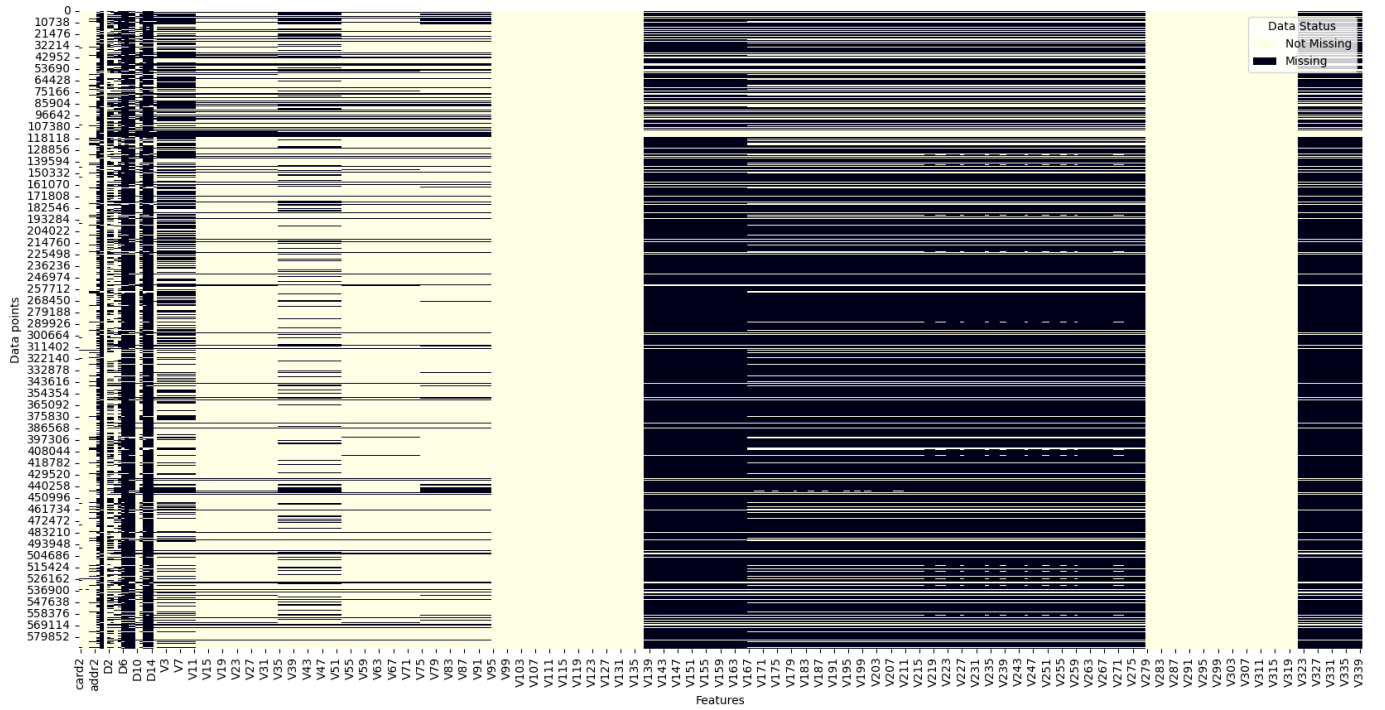


Fig. 5: Heatmap of missing values in transaction data, showing areas of data incompleteness.

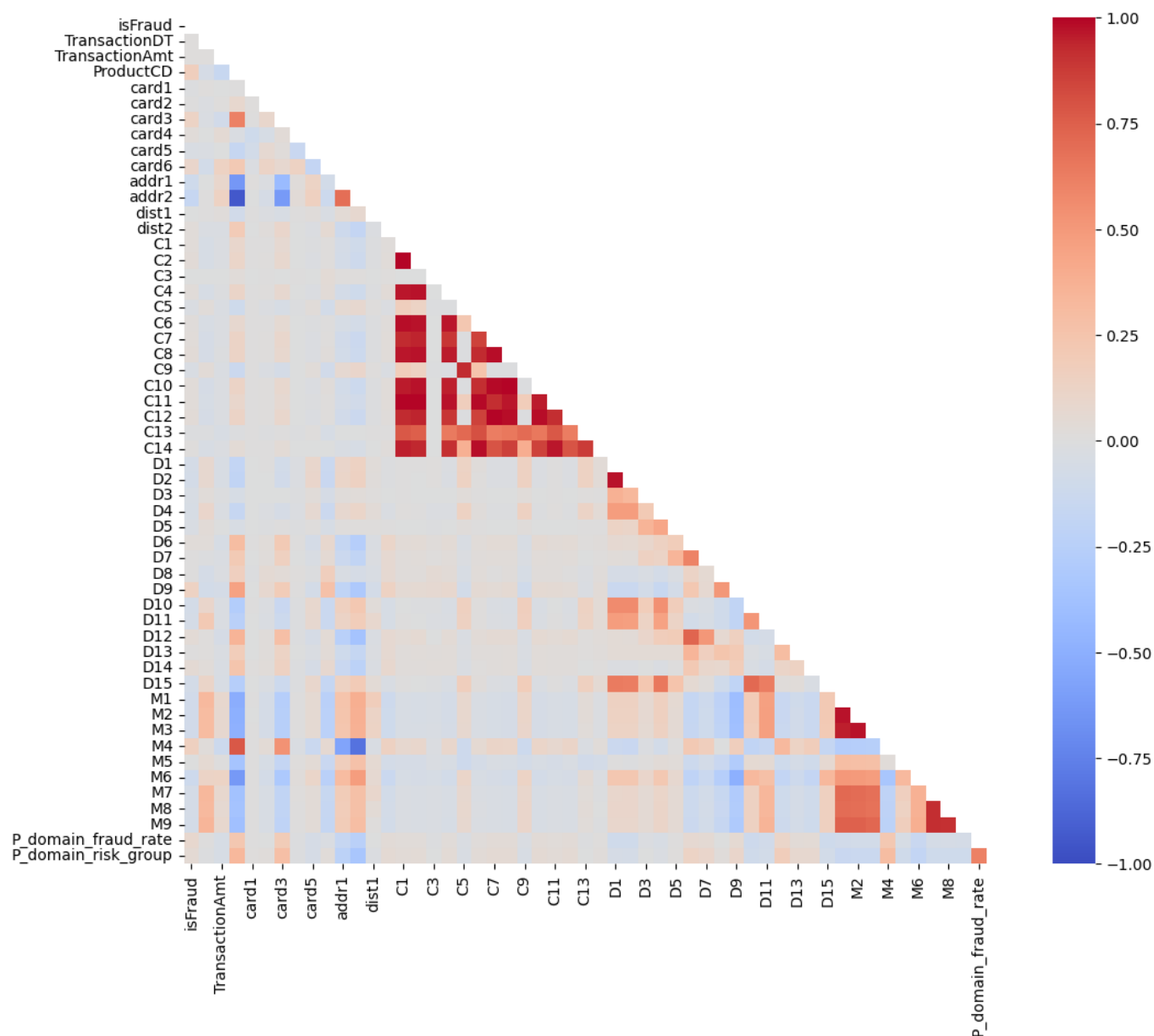


Fig. 6: Correlation heatmap of transaction features, highlighting relationships between variables.

